

Huawei | Code4Mzansi

WattWise

ENERGY GATEWAY

Smart metering, monitoring & energy-data analytics for South African households, SMMEs and municipalities.

Solving a G13 challenge: electricity inefficiencies — one of the grid issues holding back South African homes, businesses and the economy.

Team: 5iveinfinite

THE PROBLEM

Electricity Inefficiency

Monthly, estimate-based billing gives no real-time view of how power is used — causing waste, bill shock and deepening energy poverty, while municipalities and Eskom forecast demand almost blind and cover the gap with emergency diesel.

47%

of SA households are energy-poor — the poorest spend 27% of income on electricity.

40–50%

of a typical household bill is the geyser alone — the single biggest controllable load.

R21.4bn

spent by Eskom on diesel turbines in one year (FY2023) to cover peak demand.

THE SOLUTION

WattWise Energy Gateway

WattWise streams and analyses granular, per-household consumption data so homes cut waste, utilities forecast and procure energy accurately, and the grid runs on insight — not guesswork.

1

Uses existing smart meters

No rip-and-replace — ingests current smart-meter / AMI feeds.

2

Granular consumption analytics

Load disaggregation surfaces geyser, standby and anomaly waste.

3

Three role-based portals

Consumer, Municipality and Field Technician — each sees what they need.

4

Offline USSD channel

*130# reaches energy-poor homes with no data or smartphone.

WHO IT IMPACTS

Target Audience

Household Consumers

The ~47% of energy-poor SA homes plus everyday grid-connected families. They see and cut waste in real time — online via the consumer dashboard or offline over USSD *130#. Directly targets the 40–50% geyser and 15% standby loads driving their bills.

SMMEs

Small, medium & micro enterprises — spaza shops, salons, workshops, light manufacturers — where electricity is a major operating cost. Granular load visibility helps owners cut waste, avoid bill shock and stay open through load-shedding with accurate balance and outage alerts.

Municipalities & Eskom

Utilities forecast, procure, protect and maintain on evidence, not estimates: bottom-up demand forecasting shaves the peak diesel exists to cover, predictive maintenance spots failing transformers, and feeder-vs-household mismatches flag cable theft and illegal connections in near-real time.

THE IMPACT

Saving Money. Powering Government Foresight.

Real-time visibility turns wasted rands into savings for every user — and turns millions of household signals into the predictive intelligence government has never had.

SAVES

Households

Cut the 40–50% geyser & 15% standby waste in real time — money back in the pockets of homes that spend up to 27% of income on power.

SAVES

SMMEs

Lower a major operating cost: trim waste, dodge bill shock and keep trading through outages with live balance and load-shedding alerts.

SAVES

Municipalities & Eskom

Shave the peak that burned R21.4bn in diesel in one year, recover revenue lost to cable theft, and bill on data instead of estimates.

PREDICTIVE ANALYTICS FOR GOVERNMENT

Forecast demand

Bottom-up forecasts from real household load shapes — see peaks before they arrive and procure energy ahead, not in panic.

Predictive maintenance

Spot failing transformers in the data before they fail — fewer outages, lower repair and diesel costs.

Protect the grid

Feeder-vs-household mismatches flag cable theft and illegal connections in near-real time — crews dispatched before the area goes dark.

THE REVENUE MODEL

Build-Operate-Transfer

We finance, deploy and run the analytics layer on the municipality’s existing meters — they pay for outcomes, not infrastructure — then own it outright at the end of term.

1

BUILD
Integration fee

One-off setup to connect WattWise to existing AMI/ smart-meter feeds, configure portals & train staff.

R180–R450 per 1,000 meters onboarded,
billed once at deployment.

2

OPERATE
Per-meter SaaS + uptime

Recurring fee to host, analyse and support the live platform across all three portals.

R8–R14 / active meter / month,
banded by volume (see table below).

3

TRANSFER
Handover at term end

After 5–7 years the platform, data and runbooks transfer to the municipality at no extra licence cost.

Optional support retainer:
15% of final annual SaaS value.

HOW IT SCALES — PER-METER SaaS TIERS

< 50k meters
R14 / meter / mo
Pilot & small municipalities

50k – 250k
R11 / meter / mo
Mid-size metros

250k – 1M
R9 / meter / mo
Large metros

> 1M (Eskom-scale)
R8 / meter / mo
National rollout

Cost falls per meter as the base grows — revenue scales with rollout while unit price stays affordable for utilities. Indicative ZAR; finalised per tender.

OUR UNIQUE SELLING POINT

Built for Every South African

Existing home energy management systems run on their own proprietary hardware and service networks — and simply don't reach every area in South Africa. WattWise rides on infrastructure that already exists in homes nationwide.

Existing systems

- ✗ Own proprietary meters & hardware to install
- ✗ Own service network — limited geographic reach
- ✗ Maintenance gaps leave many SA areas uncovered
- ✗ Smartphone / data assumed — excludes millions

WattWise

- ✓ Integrates with existing smart meters — no new hardware
- ✓ Uses meters & networks already deployed across SA
- ✓ No coverage rollout — works wherever meters exist
- ✓ USSD *130# reaches homes with no data or smartphone

Accessible to all. By building on existing meters and USSD instead of new infrastructure, WattWise can reach the whole population — including the ~47% of energy-poor households with no smartphone or data — not just the affluent suburbs other systems can service.

WattWise

ENERGY GATEWAY

Tackling G13 electricity inefficiency for households, SMMEs and municipalities.

Questions? · Team: 5iveinfinite

SOURCES & REFERENCES

Where Our Numbers Come From

Every statistic in this deck is drawn from audited utility data, peer-reviewed research or industry sources.

~47% energy-poor · poorest spend 27% of income on power

Que et al. (2024), “Energy Poverty among Off-Grid Households,” Sustainability 16(11):4627, MDPI; corroborated by Climate Scorecard (2025).

mdpi.com/2071-1050/16/11/4627 · climatescorecard.org

40–50% of a typical household bill is the geyser

Lectric ‘n More (2025), “How Your Geyser Burns 40% of Your Power Bill”; Energy Bee (2026), 200L Geyser Price Guide (water heating 40–60%).

lectricnmore.co.za · energybee.co.za

R21.4bn spent by Eskom on diesel turbines (FY2023)

Eskom Holdings SOC Ltd — Annual Financial Statements 2023 (audited): Eskom-owned OCGT expenditure R21.4bn, up from R10bn (2022).

eskom.co.za/.../Eskom_annual_financial_statements_2023.pdf